PROBLEM VF2

Oscillating Wall under a Liquid Layer

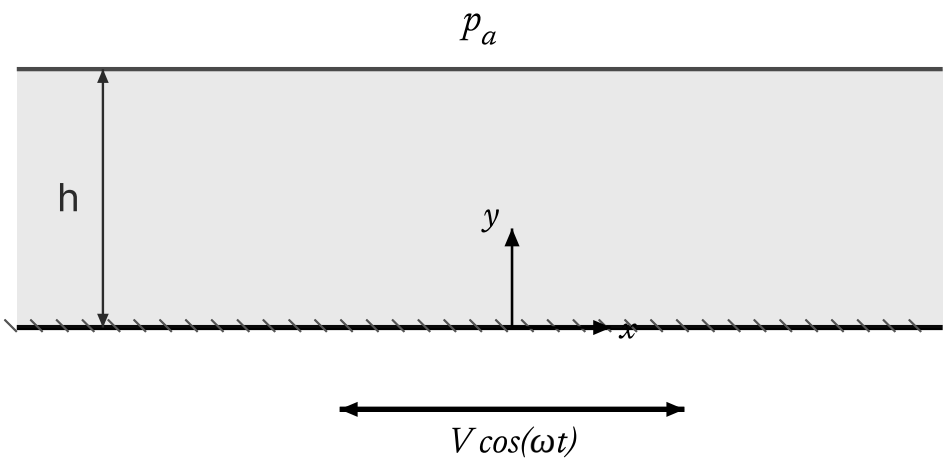
Original statement · VF2_w_solution.pdf

Unsteady unidirectional flow | Complex solution | Quasi-steady and Stokes limits

OFFICIAL PROBLEM STATEMENT

Consider the periodic motion induced in a liquid layer of thickness h when the bottom wall moves with velocity $V\cos(\omega t)$. Write the equation with boundary conditions that determines $v_x(y,t)$. Consider separately the cases $\omega h^2/\nu \gg 1$ and $\omega h^2/\nu \ll 1$. Obtain the general solution and show how it reduces to the limiting solutions studied earlier in the appriate limits.

The sheet prints "appriate"; it is kept. The top of the layer is a free surface at the pressure p_a, as the figure of the sheet shows.



Statement figure (redrawn from VF2_w_solution.pdf, p. 1): a liquid layer of thickness h with a free surface at p_a on a wall that oscillates with velocity V cos(ωt); y is normal to the wall and x along it

Source: VF2_w_solution.pdf (page 1: statement and first solution; pages 2 to 5: handwritten derivation). Notes.pdf, Chapter 7: unidirectional flow, Eq. 7.6; Stokes flow near a wall, Eqs. 7.26-7.31; oscillating wall of a channel, Eqs. 7.32-7.35.

Worked solution

Hypotheses, Data and Equation

- The liquid has constant ρ and μ. The flow is parallel to the wall: $\bar{v} = v_x(y,t)\,\bar{e}_x$, because continuity gives $\partial v_x/\partial x = 0$ (Notes.pdf, Eq. 7.1).
- **Pressure.** The y component of the momentum equation gives $0 = -\partial P/\partial y$ with $P = p + \rho g y$, so P is the same at every height. At the free surface $p = p_a$ and $y = h$, hence $P = p_a + \rho g h$, a constant: $\partial P/\partial x = 0$ and $P_l = 0$ (the free surface stays flat).
- **Free surface.** The air above exerts a negligible shear stress (its viscosity is neglected), so the shear stress of the liquid vanishes at $y = h$: $\partial v_x/\partial y = 0$.
- **Periodic solution.** The wall velocity is periodic and the sought solution is the permanent one, so no initial condition is needed (Notes.pdf, p. 78). A start from rest ($v_x = 0$ at $t = 0$) only matches the wall condition: because of the phase lag the periodic solution has $v_x(y,0) \neq 0$ for $y > 0$, so the start-up transient, which dies out after a time of order h^2/ν, has to be added; afterwards the periodic solution is all that remains and no initial condition is needed.

Equation (7.6) with $P_l = 0$ and $\nu = \mu/\rho$, and its boundary conditions:

$$\frac{\partial v_x}{\partial t} = \nu \frac{\partial^2 v_x}{\partial y^2}; \qquad y = 0 : \; v_x = V\cos\omega t, \qquad y = h : \; \frac{\partial v_x}{\partial y} = 0$$

Symbol	Meaning	SI unit
h	thickness of the layer	m
V, ω	amplitude and frequency of the wall velocity	m/s, s^{-1}
ν	kinematic viscosity	m^2/s
$\omega h^2/\nu$	ratio of the viscous time h^2/ν to ω^{-1}	1

General Solution with a Complex Function

Since $V\cos\omega t = \mathrm{Re}\left(Ve^{\imath\omega t}\right)$, the linear problem is solved for a complex v^* with $v_x = \mathrm{Re}(v^*)$:

$$\frac{\partial v^*}{\partial t} = \nu\frac{\partial^2 v^*}{\partial y^2}; \qquad y=0:\ v^* = Ve^{\imath\omega t}, \qquad y=h:\ \frac{\partial v^*}{\partial y} = 0$$

Look for $v^* = Ve^{\imath\omega t}f(y)$. Substituting, $\imath\omega f = \nu f''$, that is

$$f'' - \frac{\imath\omega}{\nu}f = 0; \qquad f(0) = 1, \qquad f'(h) = 0$$

(in the key's non-dimensional form, $z = y/h$ and $2\Omega^2 = \omega h^2/\nu$, the problem has **one parameter**: $f'' - 2\imath\Omega^2 f = 0$ with $f(0)=1$, $f'(1)=0$).

Characteristic equation. $\lambda^2 = \imath\omega/\nu$. With $\imath = e^{\imath\pi/4}\cdot e^{\imath\pi/4}$, $\sqrt{\imath} = (1+\imath)/\sqrt{2}$, so the two roots are $\pm\lambda$ with

$$\lambda = (1+\imath)\sqrt{\frac{\omega}{2\nu}}, \qquad \lambda h = (1+\imath)\,\Omega \quad \text{with} \quad \Omega = \sqrt{\frac{\omega h^2}{2\nu}}$$

(check: $\lambda^2 = (1+\imath)^2\,\omega/(2\nu) = 2\imath\,\omega/(2\nu) = \imath\omega/\nu$).

Constants. $f = C_1e^{\lambda y} + C_2e^{-\lambda y}$. The wall condition gives $C_1 + C_2 = 1$. The condition at the free surface, $f'(h) = \lambda\left(C_1e^{\lambda h} - C_2e^{-\lambda h}\right) = 0$, gives $C_1e^{\lambda h} = C_2e^{-\lambda h}$. Substituting $C_2 = 1 - C_1$:

$$C_1\left(e^{\lambda h} + e^{-\lambda h}\right) = e^{-\lambda h} \implies C_1 = \frac{e^{-\lambda h}}{e^{\lambda h}+e^{-\lambda h}}, \qquad C_2 = 1 - C_1 = \frac{e^{\lambda h}}{e^{\lambda h}+e^{-\lambda h}}$$

Putting them in f:

$$f = \frac{e^{-\lambda h}e^{\lambda y} + e^{\lambda h}e^{-\lambda y}}{e^{\lambda h} + e^{-\lambda h}} = \frac{e^{\lambda(h-y)} + e^{-\lambda(h-y)}}{e^{\lambda h} + e^{-\lambda h}} = \frac{\cosh\left[\lambda(h-y)\right]}{\cosh(\lambda h)}$$

and the complex solution is

$$v^* = V e^{\imath\omega t}\,\frac{\cosh\left[(1+\imath)\Omega(1-y/h)\right]}{\cosh\left[(1+\imath)\Omega\right]}, \qquad v_x = \mathrm{Re}(v^*)$$

Discrepancy with the official solution

The second boundary-condition equation of the key, boxed on page 1, reads $C_1 e^{(1+\imath)\sqrt{\omega/2\nu}\,h} - C_2 e^{-(1+\imath)\sqrt{\omega/2\nu}\,h} = 1$. The condition $f'(h) = 0$ gives 0 on the right-hand side, as derived above, and the coefficients C_1, C_2 that the key writes just below are the ones that follow from 0, not from 1. It is a slip in the key; the final solution is not affected.

The Real Solution

With $s = \Omega(1-y/h)$ and $\cosh(s+\imath s) = \cosh s\cos s + \imath\sinh s\sin s$:

$$f = \frac{A + \imath B}{C + \imath D}, \qquad A = \cosh s\cos s,\ B = \sinh s\sin s,\ C = \cosh\Omega\cos\Omega,\ D = \sinh\Omega\sin\Omega$$

Multiplying numerator and denominator by $C - \imath D$ gives $f = \left[(AC+BD) + \imath(BC-AD)\right]/(C^2+D^2)$. The real part of $e^{\imath\omega t}f$ is $\left[(AC+BD)\cos\omega t - (BC-AD)\sin\omega t\right]/(C^2+D^2)$ (since $\mathrm{Re}[(\cos\omega t + \imath\sin\omega t)(a+\imath b)] = a\cos\omega t - b\sin\omega t$). Also $C^2+D^2 = (\cosh 2\Omega + \cos 2\Omega)/2$, which is the denominator $e^{2\Omega} + e^{-2\Omega} + 2\cos 2\Omega$ of the key up to the factor 4. Then

$$\frac{v_x}{V} = \frac{(AC+BD)\cos\omega t - (BC-AD)\sin\omega t}{C^2+D^2}$$

At $y = 0$, $s = \Omega$, so $A = C$ and $B = D$ and $v_x/V = \cos\omega t$, as the wall condition requires. The key obtains the same function as a sum of four terms of the form $e^{\pm\Omega z}\cos(\Omega z \pm \omega t + \dots)$, multiplying by the conjugate of the denominator.

Limit $\omega h^2/\nu \ll 1$: Quasi-steady Solution

Here $|\lambda h| = \sqrt{2}\,\Omega \ll 1$, so the viscous time h^2/ν is much shorter than ω^{-1} and the unsteady term is negligible. The equation reduces to $\partial^2 v_x/\partial y^2 = 0$ with $v_x = V\cos\omega t$ at $y = 0$ and $\partial v_x/\partial y = 0$ at $y = h$. Integrating twice, $v_x = A(t)y + B(t)$; the condition at $y = h$ gives $A = 0$ and the wall gives $B = V\cos\omega t$:

$$v_x = V\cos\omega t$$

The liquid layer moves like a solid: with no shear stress at the free surface, nothing opposes the motion. The exact solution gives the same: $\cosh(x) \simeq 1 + x^2/2$ for small x, so $f \simeq 1 + \frac{\lambda^2}{2}\left[(h-y)^2 - h^2\right] = 1 - \frac{\omega}{2\nu}\,y\,(2h - y) \to 1$. The first correction (computed here, an **extension**) is $v_x \simeq V\left[\cos\omega t + \frac{\omega}{2\nu}\,y\,(2h-y)\sin\omega t\right]$, of relative size $\omega h^2/\nu \ll 1$ (it uses $\mathrm{Re}\left[\imath e^{\imath\omega t}\right] = -\sin\omega t$).

Limit $\omega h^2/\nu \gg 1$: Stokes Layer

Here $\Omega \gg 1$: the thickness h is much larger than the penetration length $\delta = \sqrt{\nu/\omega}$. The sheet first writes $\partial v_x/\partial t = 0$ with two question marks: the right argument is not that the layer is at rest, but that the motion is confined to a thin layer. Balancing $\omega V \sim \nu V/\delta^2$ gives $\delta \sim \sqrt{\nu/\omega} \ll h$, and the problem becomes the one of Stokes (Notes.pdf, Eq. 7.26):

$$\frac{\partial v_x}{\partial t} = \nu\frac{\partial^2 v_x}{\partial y^2}; \qquad y = 0:\ v_x = V\cos\omega t, \qquad y \gg \sqrt{\nu/\omega}:\ v_x \to 0$$

whose solution (7.31) is

$$v_x = V\, e^{-y/\sqrt{2\nu/\omega}} \cos\left(\omega t - \frac{y}{\sqrt{2\nu/\omega}}\right)$$

It follows from the exact solution: $\cosh[\lambda(h-y)]/\cosh(\lambda h) = \left[e^{-\lambda y} + e^{-\lambda(2h-y)}\right] / \left[1 + e^{-2\lambda h}\right]$, and $\left|e^{-2\lambda h}\right| = e^{-2\Omega} \to 0$, $\left|e^{-\lambda(2h-y)}\right| \le e^{-\Omega}$ for $y \le h$, so $f \simeq e^{-\lambda y}$ with $\lambda = (1+\imath)/\sqrt{2\nu/\omega}$, and the real part of $V e^{\imath\omega t} e^{-\lambda y}$ is the expression above. The thickness h **disappears** from the result.

The key solves the same limit in the non-dimensional variable $\eta = y/\delta$ with $\delta^2\omega/\nu = 1$: $f'' - \imath f = 0$, $f(0) = 1$, $f(\infty)$ bounded, so the root with growing exponential has a zero coefficient ($A = 0$, $B = 1$) and $f = e^{-(1+\imath)\eta/\sqrt{2}}$.

Answers and Checks

- Problem: $\partial v_x/\partial t = \nu\, \partial^2 v_x/\partial y^2$, $v_x(0,t) = V\cos\omega t$, $\partial v_x/\partial y(h,t) = 0$.
- General solution: $v_x = \mathrm{Re}\left\{V e^{\imath\omega t} \cosh\left[(1+\imath)\Omega(1-y/h)\right] / \cosh\left[(1+\imath)\Omega\right]\right\}$ with $\Omega^2 = \omega h^2/(2\nu)$.
- $\omega h^2/\nu \ll 1$: $v_x = V\cos\omega t$ (rigid motion). $\omega h^2/\nu \gg 1$: Stokes solution, independent of h.
- Boundary conditions of the general solution: at $y = 0$, $f = 1$; at $y = h$, $\sinh[\lambda(h-y)] = 0$, so $f' = 0$.
- Dimensions: $\omega h^2/\nu$ is $T^{-1}L^2/(L^2T^{-1}) = 1$; $\sqrt{2\nu/\omega}$ is a length.
- Check against the official key: VF2_w_solution.pdf gives the same equation, complex solution, real part, rigid limit and Stokes limit, with the slip in the second boundary-condition equation noted above; the minus sign that the key writes before $e^{-\eta/\sqrt{2}} e^{\imath(\omega t - \eta/\sqrt{2})}$ on page 4 (a reading of the handwriting) also disappears in the real part it boxes.